

```

clc;
clear;
close all
% Specifications
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 1000;     % Passband Frequency (Hz)
Fst = 1500;    % Stopband Frequency (Hz)
% Butterworth Filter Design
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
[b, a] = butter(N, Wn);
% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');
% Impulse Response
figure;
[h, n] = impz(b, a, 50);
stem(n, h, 'filled');
title('Impulse Response');
xlabel('Samples');
ylabel('Amplitude');
grid on;
% Step Response
figure;
stepz(b, a);
title('Step Response');
grid on;
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');

```

% Display Results

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

```
printf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```



